

Math 74: Algebraic Topology

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Problem 1. Show that if $h, h' : X \rightarrow Y$ are homotopic and $k, k' : Y \rightarrow Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.

Solution.

Assume $h \simeq h'$ and $k \simeq k'$. Then there exist F_h and F_k such that

$$F_h(x, 0) = h(x), F_h(x, 1) = h'(x), F_k(x, 0) = k(x), F_k(x, 1) = k'(x)$$

To show that $k \circ h \simeq k' \circ h'$, we define $F : X \times I \rightarrow Z$ as,

$$F(x, t) = F_k(F_h(x, t), t)$$

Note that F is the composition of continuous maps and hence is continuous. We verify initial and final points.

$$F(x, 0) = F_k(F_h(x, 0), 0) = F_k(h(x), 0) = k(h(x)) = k \circ h(x)$$

$$F(x, 1) = F_k(F_h(x, 1), 1) = F_k(h'(x), 1) = k'(h'(x)) = k' \circ h'(x)$$

Therefore there exists a continuous deformation of $k \circ h$ to $k' \circ h'$, and $k \circ h \simeq k' \circ h'$.

Problem 2. A space X is said to be contractible if the identity map $i_X : X \rightarrow X$ is homotopic to a constant map.

1. Show that I and $\mathbb{R}$ are contractible.
2. Show that a contractible space is path connected.
3. Show that if Y is contractible, then for any X the set $[X, Y]$ has a single element.
4. Show that if X is contractible and Y is path connected, then $[X, Y]$ has a single element.

Solution.

1. Let $i : I \rightarrow I$, $i(x) = x$ be the identity map, and wlog let $e_0 : I \rightarrow I$, $e_0(x) = 0$ be the constant map at zero. Then we define a map $H : I \times I \rightarrow I$ as follows,

$$H(x, t) = tx$$

Note that H is continuous (it is a polynomial) and for all $x, t \in [0, 1]$ we have $tx \in [0, 1]$. Moreover

$$H(x, 0) = 0 = e_0(x) \quad \text{and} \quad H(x, 1) = x = i(x)$$

Hence $e_0 \simeq i$ and I is contractible.

Similarly, let $i : \mathbb{R} \rightarrow \mathbb{R}$, $i(x) = x$ be the identity map, and wlog let $e_0 : \mathbb{R} \rightarrow \mathbb{R}$, $e_0(x) = 0$ be the constant map at zero. Then we define a map $H : \mathbb{R} \times I \rightarrow \mathbb{R}$ as follows,

$$H(x, t) = tx$$

Note that H is continuous (it is a polynomial), and we have

$$H(x, 0) = 0 = e_0(x) \quad \text{and} \quad H(x, 1) = x = i(x)$$

Hence $e_0 \simeq i$ and $\mathbb{R}$ is contractible.

2. Let X be a contractible space. Then for any two points $a, b \in X$ we know that $i_x \simeq e_a$ and $i_x \simeq e_b$. By transitivity of homotopy, we have $e_a \simeq e_b$ and there exists a continuous map $H(s, t)$ such that $H(x, 0) = e_a$ and $H(x, 1) = e_b$ for all $x \in X$.

Now fix any point x_0 and define the map $\varphi : I \rightarrow X$, where $\varphi(t) = H(x_0, t)$. Note that φ is continuous because it is a restriction of H and moreover $\varphi(0) = a$ and $\varphi(1) = b$. Since a, b were arbitrary points in X this holds for any two points and hence X is path-connected.

3. Assume Y is contractible. Let X be another topological space, and $f, g : X \rightarrow Y$ be continuous maps. Let c be some point in Y . Define a map $h : Y \rightarrow Y$ where $h(y) = c$, for all $y \in Y$. Note that,

$$h \circ f = h \circ g \implies h \circ f \simeq h \circ g$$

Since Y is contractible, by definition we have $\text{id}_Y \simeq h$. Then,

$$\text{id}_Y \circ f \simeq \text{id}_Y \circ g \implies f \simeq g$$

Since f, g were arbitrary maps this implies that any two continuous maps from X to Y are homotopic. Thus $||X, Y|| = 1$.

4. Assume X is contractible and Y is path-connected. Let $h : X \rightarrow X$ such that $h(x) = c$ for some constant point $c \in X$. Also let $f, g : X \rightarrow Y$ two continuous functions.

$$X \xrightarrow{h} X \begin{matrix} \xrightarrow{f} \\ \xrightarrow{g} \end{matrix} Y$$

Then, consider $g \circ h, f \circ h : X \rightarrow Y$. This gives us two points in Y , say $f(c) = a$ and $g(c) = b$ respectively. Since Y is path connected there exists some function $p : I \rightarrow Y$ where $p(0) = a, p(1) = b$. Then we define a homotopy between $g \circ h$ and $f \circ h$ by $H : X \times I \rightarrow Y$,

$$H(x, t) = p(t)$$

Then $H(x, 0)$ gives us,

$$H(x, 0) = p(0) = a = f(c)$$

and similarly,

$$H(x, 1) = p(1) = b = g(c)$$

Thus,

$$g \circ h \simeq f \circ h$$

But since X is contractible we have $h \simeq \text{id}_X$. So, we have

$$g \circ \text{id}_X \simeq f \circ \text{id}_X \implies g \simeq f$$

Since f, g were arbitrary maps this implies that any two continuous maps from X to Y are homotopic. Thus $||X, Y|| = 1$.

Problem 3. A subset A of $\mathbb{R}^n$ is said to be star convex if for some point $\mathbf{a}_0$ of A , all the line segments joining $\mathbf{a}_0$ to other points of A lie in A .

1. Find a star convex set that is not convex.
2. Show that if A is star convex, A is simply connected.
3. Show that if A is star convex, any two paths in A having the same initial and final points are path homotopic.

Solution.

1. $(x, 0) \cup (0, y) \in \mathbb{R}^2$ is star convex at $(0, 0)$ but not convex.
2. Assume A is star convex. Let $\mathbf{a}_0$ be a point of A that has line segments joining it to all other points of A . Note that by definition A is path-connected. Then we only need to show that $\pi_1(A, \mathbf{a}_0) \cong \{0\}$.

Let p, q be loops based at $\mathbf{a}_0$. Then we have the map F defined as follows:

$$F(x, t) = (1 - t)p(x) + t\mathbf{a}_0$$

where because A is star convex with center $\mathbf{a}_0$ the line segment joining $p(x)$ to $\mathbf{a}_0$ lies in A , so $F(x, t) \in A$ for all $x, t \in I \times I$. Now note that F is a path homotopy between p and the constant function $e_{\mathbf{a}_0}$ since

$$F(x, 0) = p(x), \quad F(x, 1) = e_{\mathbf{a}_0}(x)$$

And

$$F(0, t) = F(1, t) = \mathbf{a}_0$$

Similarly, we define the map F' ,

$$F'(x, t) = (1 - t)q(x) + t\mathbf{a}_0$$

which is a path homotopy between q and the constant function $e_{\mathbf{a}_0}$ since

$$F'(x, 0) = q(x), \quad F'(x, 1) = e_{\mathbf{a}_0}(x)$$

(again, since A is star convex with center $\mathbf{a}_0$ the line segment joining $q(x)$ to $\mathbf{a}_0$ lies in A), and

$$F(0, t) = F(1, t) = \mathbf{a}_0$$

By transitivity of path-homotopies we have,

$$p \simeq e_{\mathbf{a}_0} \simeq q \implies p \simeq q$$

Since p, q were arbitrary loops based at $\mathbf{a}_0$ this holds for any two loops and we have $\pi_1(A, \mathbf{a}_0) \cong \{0\}$. Hence, a star convex space is simply connected.

3. From part (2) we know that if A is star convex, it is simply connected. And Lemma 52.3 states that in any simply connected space, two paths having the same initial and final points are path homotopic.

Problem 4. Let x_0 and x_1 be two given points of the path-connected space X . Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1 , we have $\widehat{\alpha} = \widehat{\beta}$.

Solution.

($\implies$)

Assume $\pi_1(X, x_0)$ is abelian. Note that $[\alpha \star \bar{\beta}] \in \pi_1(X, x_0)$. For any $[f] \in \pi_1(X, x_0)$ we have

$$\begin{aligned} [f] \star [\alpha \star \bar{\beta}] &= [\alpha \star \bar{\beta}] \star [f] \\ [f] \star [\alpha] \star [\bar{\beta}] &= [\alpha] \star [\bar{\beta}] \star [f] \\ [\bar{\alpha}] \star [f] \star [\alpha] \star [\bar{\beta}] \star [\beta] &= [\bar{\alpha}] \star [\alpha] \star [\bar{\beta}] \star [f] \star [\beta] \\ [\bar{\alpha}] \star [f] \star [\alpha] &= [\bar{\beta}] \star [f] \star [\beta] \\ \widehat{\alpha}([f]) &= \widehat{\beta}([f]) \end{aligned}$$

Since α, β , and f were arbitrary choices this implies that for every pair α and β of paths from x_0 to x_1 , we have $\widehat{\alpha} = \widehat{\beta}$.

($\impliedby$)

Assume for every pair α, β of paths from x_0 to x_1 we have $\widehat{\alpha} = \widehat{\beta}$. In particular when $x_1 = x_0$ this holds for any pair of curves $[\gamma], [\delta]$ based at x_0 . Then,

$$\begin{aligned} \widehat{\gamma}([\gamma]) &= \widehat{\delta}([\gamma]) \\ [\bar{\gamma}] \star [\gamma] \star [\gamma] &= [\bar{\delta}] \star [\gamma] \star [\delta] \\ [\delta] \star [\gamma] &= [\gamma] \star [\delta] \end{aligned}$$

Hence $\pi_1(X, x_0)$ is abelian.

Problem 5. Let A be a subset of $\mathbb{R}^n$ let $h : (A, a_0) \rightarrow (Y, y_0)$. Show that if h is extendable to a continuous map of $\mathbb{R}^n$ into Y , then h_* is the zero homomorphism (the trivial homomorphism that maps everything to the identity element).

Solution.

Let $i : A \hookrightarrow \mathbb{R}^n$. We also know that there exists a continuous map $\tilde{h} : \mathbb{R}^n \rightarrow Y$, such that $\tilde{h}|_A = h$. Then,

$$h = \tilde{h} \circ i$$

And over fundamental groups this implies,

$$h_* = \tilde{h}_* \circ i_*$$

But from class we know that $\pi_1(\mathbb{R}^n, a_0) \cong \{e\}$. So, i_* must send every element to the identity. Then for any $[f] \in \pi_1(A, a_0)$

$$h_*([f]) = \tilde{h}_*(i_*([f])) = \tilde{h}_*(e) = e$$

Therefore $h_* = \tilde{h}_* \circ i_*$ is the trivial homomorphism.

Problem 6. Let G be a topological group with operation $\cdot$ and the identity element x_0 . Let $\Omega(G, x_0)$ denote the set of all loops in G based at x_0 . If $f, g \in \Omega(G, x_0)$, let us define a loop $f \otimes g$ by the rule

$$(f \otimes g)(s) = f(s) \cdot g(s)$$

1. Show that this operation makes the set $\Omega(G, x_0)$ into a group.
2. Show that this operation induces a group operation $\otimes$ on $\pi_1(G, x_0)$.
3. Show that the two group operations $\star$ and $\otimes$ on $\pi_1(G, x_0)$ are the same. [*Hint:* Compute $(f \star e_{x_0}) \otimes (e_{x_0} \star g)$].
4. Show that $\pi_1(G, x_0)$ is abelian.

Solution.

1. Let $f, g \in \Omega(G, x_0)$.

- Identity: The constant function e_{x_0} is the identity.

$$(f \otimes e_{x_0})(s) = f(s) \cdot e_{x_0}(s) = f(s) = e_{x_0}(s) \cdot f(s) = (e_{x_0} \otimes f)(s)$$

- Associativity: For $f, g, h \in \Omega(G, x_0)$, $s \in I$, we have

$$((f \otimes g) \otimes h)(s) = (f(s) \cdot g(s)) \cdot h(s) = f(s) \cdot (g(s) \cdot h(s)) = (f \otimes (g \otimes h))(s),$$

where associativity follows from associativity of the group operation in G .

- Closure under inverses: for any given loop f we can define its inverse as

$$f^{-1}(x) = (f(x))^{-1}$$

- Closure under the operation: since $(f \otimes g)(0) = f(0) \cdot g(0) = x_0$ and $(f \otimes g)(1) = f(1) \cdot g(1) = x_0$ the group is closed under $\otimes$.

2. Let $[f], [g] \in \pi_1(G, x_0)$. We only need to show that $\otimes$ is well-defined on homotopy classes, because the other group axioms follow from (1). Let $f \simeq f'$ and $g \simeq g'$. We want to show that $f \otimes g \simeq f' \otimes g'$.

Then we have homotopies $P, Q : I \times I \rightarrow G$ where

$$P(x, 0) = f(x), P(x, 1) = f'(x)$$

and

$$Q(x, 0) = g(x), Q(x, 1) = g'(x)$$

Then we define a map $H : I \times I \rightarrow G$,

$$H(x, t) = P(x, t) \cdot Q(x, t)$$

Note that this is a path homotopy between $f \otimes g$ and $f' \otimes g'$ because the initial and final points match,

$$H(0, t) = H(1, t) = x_0 \cdot x_0 = x_0$$

$$H(x, 0) = f \otimes g,$$

$$H(x, 0) = P(x, 0) \cdot Q(x, 0) = f(x) \cdot g(x)$$

$$\text{and } H(x, 1) = f' \otimes g',$$

$$H(x, 1) = P(x, 1) \cdot Q(x, 1) = f'(x) \cdot g'(x)$$

Hence, $f \otimes g \simeq f' \otimes g'$, and thus the operation induces a group operation on $\pi_1(G, x_0)$.

3. Note that since $f \simeq f \star e_{x_0}$ and $g \simeq e_{x_0} \star g$ we have,

$$f \otimes g = (f \star e_{x_0}) \otimes (e_{x_0} \star g)(s) = \begin{cases} f(2s), & s \in [0, \frac{1}{2}] \\ g(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases} = f \star g$$

4. From (3) we know that $(f \star e_{x_0}) \otimes (e_{x_0} \star g)(s) = f \star g$ and that $g \star f = g \otimes f$. Now observe that,

$$g \otimes f = (e_{x_0} \star g) \otimes (f \star e_{x_0})(s) = \begin{cases} f(2s), & s \in [0, \frac{1}{2}] \\ g(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases} = f \star g$$

That is, $f \star g = g \star f$. Since f, g were arbitrary loops in $\pi_1(G, x_0)$, the fundamental group is abelian.

Problem 7. Show that the map $f : S^1 \rightarrow S^1$, defined as $f(z) = z^n$ for $z \in S^1 \subset \mathbb{C}$, is a covering map. Here we think of S^1 as $\{z \in \mathbb{C}, \text{ such that } |z| = 1\}$.

Solution.

Note that $z = e^{i\theta}$ for all $z \in S^1$. Then we have,

$$f(z) = f(e^{i\theta}) = e^{ni\theta}$$

We will construct an evenly covered neighborhood U for every point $z_0 \in S^1$. Let $z_0 = e^{i\phi}$.

Pick $\varepsilon = \pi/n$, and define an open neighborhood of z_0

$$U := (e^{i(\phi-\varepsilon)}, e^{i(\phi+\varepsilon)})$$

Note that $f(e^{i\theta}) \in U$ if and only if $e^{in\theta} = e^{i\alpha}$ for some $\alpha \in (\phi - \varepsilon, \phi + \varepsilon)$. But since $e^{i\theta}$ is 2π periodic this is equivalent to,

$$f(e^{i\theta}) \in U \iff n\theta \in (\phi - \varepsilon + 2\pi k, \phi + \varepsilon + 2\pi k)$$

for some $k \in \mathbb{Z}$. Dividing by n gives us,

$$\theta \in \left(\frac{\phi - \varepsilon}{n} + \frac{2\pi k}{n}, \frac{\phi + \varepsilon}{n} + \frac{2\pi k}{n} \right).$$

Because $2\varepsilon < \frac{2\pi}{n}$, these intervals are pairwise disjoint modulo 2π , and there are exactly n distinct choices of $k = 0, 1, \dots, n-1$ (Since for $k = n$, we have $\theta + 2\pi k/n \equiv \theta + 2\pi \equiv \theta$). Now for each k we define,

$$V_k := \left\{ e^{i\theta} : \theta \in \left(\frac{\phi - \varepsilon}{n} + \frac{2\pi k}{n}, \frac{\phi + \varepsilon}{n} + \frac{2\pi k}{n} \right) \right\}.$$

where V_k is open in S^1 , the sets V_k are disjoint, and

$$f^{-1}(U) = \bigsqcup_{k=0}^{n-1} V_k.$$

Now for each k the restriction

$$f|_{V_k} : V_k \rightarrow U$$

is a homeomorphism. Note that it is continuous (because it is the restriction of a continuous function), bijective, and its inverse is given by

$$e^{i\alpha} \mapsto e^{i(\alpha + 2\pi k)/n}$$

which is continuous on U because U lies in an arc of length less than 2π . Thus every point $z_0 \in S^1$ has an evenly covered neighborhood, and f is a covering map.